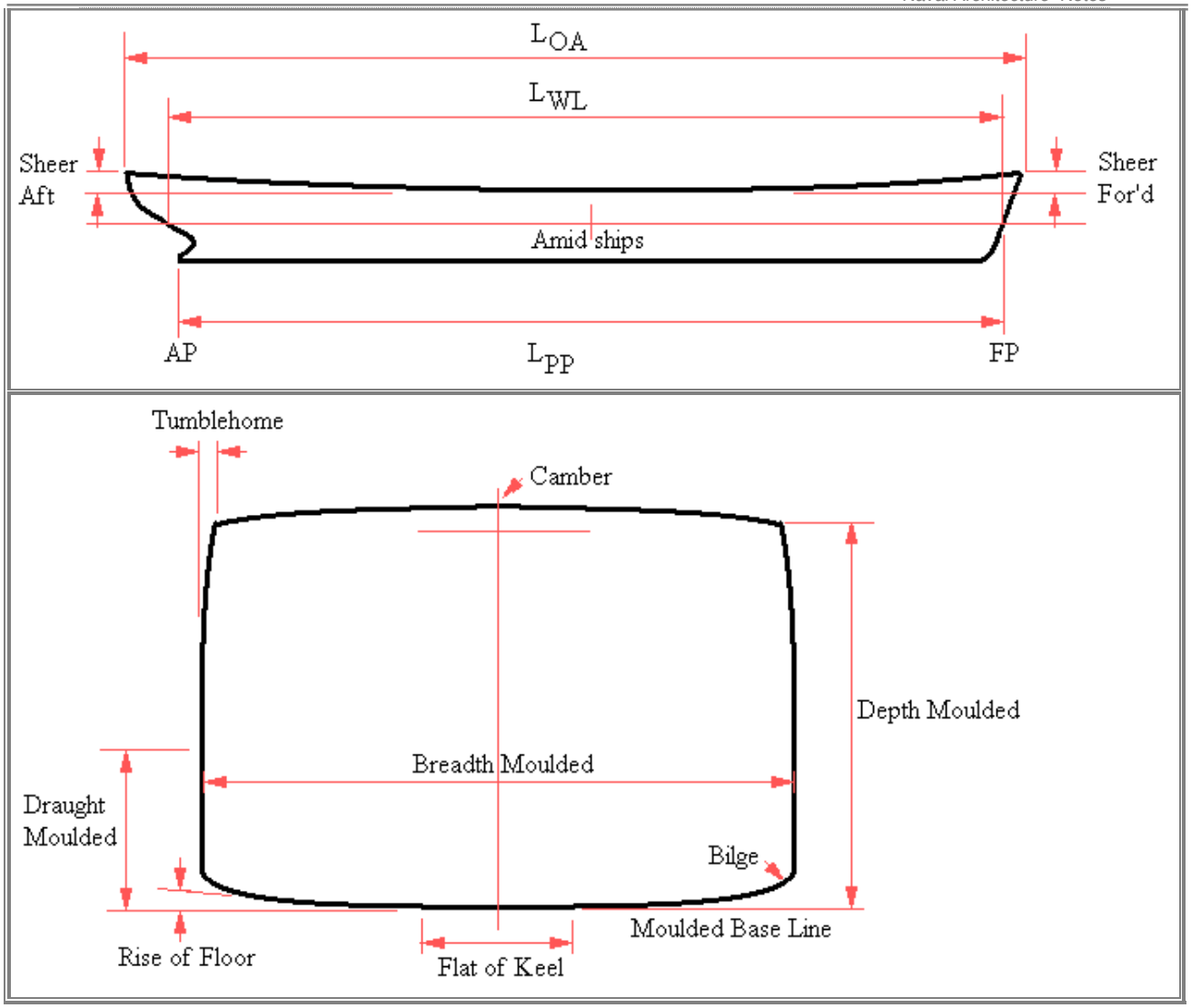
Basic Terms, Terminologies and Symbols

After perpendicular(AP)	This is represented by a line which is perpendicular to the intersection of the after edge of the rudder-post with the designed load water-line. This is the case for both single- and twin-screw merchant ships. For some classes of warships, and for merchant ships having no rudder-post, the after perpendicular is taken as the centre-line of the rudder stock.
Amidships (⋈)	This is the point midway between the forward and after perpendiculars.
Breadth (B)	This is the maximum beam, or breadth, of the ship measured at amidships.
Bilge	This is the rounded plating at the lower corners between the vertical shell plating and the outer bottom plating.
BM_L	Longitudinal metacentric radius measured from centre of buoyancy
BM_T	Transverse metacentric radius measured from centre of buoyancy
Block coefficient (C_B)	This is a measure of the fullness of the form of the ship and is the ratio of the volume of displacement to a given water-line, and the volume of the circumscribing solid of constant rectangular cross-section having the same length, breadth and draught as the ship. ie: $C_B = \nabla \div (L \times B \times T)$ The L_{PP} is normally used in calculating the value of C_B which varies with the type of ship. Fast ships 0.50-0.65 (fine form) Ordinary ships 0.65-0.75 (moderate form) Slow ships 0.75-0.85 (full form)
Camber or round of beam	This is the transverse curvature given to the decks, and is measured by the difference between the heights of the deck at side and centre. The amount of camber amidships is often one-fiftieth of the beam of the ship.
Coefficients of form	Form is used as a general term to describe the shape of the ship's hull; and when comparing one ship's form with another, the naval architect makes use of a number of coefficients. These coefficients are of great use in power, stability, strength and design calculations. Examples are Cb, Cp, Cw etc.
Centre of flotation (F)	This is the centre of the area, or centroid, of the water-plane of a ship. For small angles of trim consecutive water-lines pass through F. The location is normally on the centerline and longitudinally the distance from AP or amidships is referred to as LCF
Centre of buoyancy (B)	This is the centroid of the underwater form of a ship, and is the point through which the total force of buoyancy may be assumed to act. Its position is defined by: (a) KB the vertical distance above the base, sometimes referred to as VCB (b) LCB the longitudinal distance measured either from amidships or AP or FP.
Centre of gravity (G)	This is the point through which the total weight of the ship may be assumed to act. It also is defined by: (a) KG the vertical distance above the base (b) LCG the longitudinal distance measured either from amidships or AP or FP

C_P	Prismatic coefficient, $C_P = \frac{\nabla}{A_M \times L}$
Depth (D)	This is the vertical distance between the base line and the top of the uppermost continuous deck measured at the side amidships.
Draught (T)	This is the depth of immersion from the keel to any waterline.
Displacement	This equals the volume (∇) or weight (Δ) of water displaced by the hull.
Displacement as a volume (∇)	This is volume of water displaced by the ship. It can be imagined as the volume of the hole in the water occupied by the ship measured in cubic metres.
Displacement as a weight (Δ)	This is the weight of water displaced by the ship. It equals the volume displaced multiplied by a constant representing the density of water, ie: In fresh water $\Delta = \nabla$ x 1000 kg/m³ In sea water $\Delta = \nabla$ x 1025 kg/m³ Weight (or mass) displacement equals the total weight of the ship when the ship is at rest in equilibrium in still water.
Deadweight	This is the difference between the weight displacement and the lightship weight. This is the measure of a ship's capacity to carry cargo, fuel, passengers, stores, etc, expressed in tonnes. The size of tankers is often given in terms of deadweight tonnage, which is the design deadweight. Ships are usually chartered on the deadweight tonnage.
Displacement tonnage	This represents the designed total weight of the ship. It is the sum of lightship weight and deadweight. The size of warshipsand government ships is always given in terms of displacement tonnage.
Entrance and run	These are the shaped underwater portions of the ship forward and aft of the parallel middle body.
Forward perpendicular (FP)	This is represented by a line which is perpendicular to the intersection of the designed load water-line with the forward side of the stem.
Freeboard	This may be considered to be the height amidships, of the freeboard deck at side above the normal summer load water-line.
GM_L	Longitudinal metacentric height measured from centre of gravity
GM_T	Transverse metacentric height measured from centre of gravity
Gross tonnage (GRT)	This is a measure of the total volume of enclosed spaces in a ship including the under-deck, 'tween-deck spaces and enclosed spaces above the upper deck. The size of most ordinary merchant ships is quoted in terms of gross tonnage. Although it unit is tons, it must be remembered that it is a measure of volume, not weight. 1 ton = 100 ft3.
Heel (∅)	This is the amount of inclination of the ship in the transverse direction, and is usually measured in degrees.
I_L	Longitudinal moment of inertia of waterplane about amidship
I_{LCF}	Longitudinal moment of inertia of waterplane about F
I_T	Transverse moment of inertia of waterplane about centreline
KM_L	Height of longitudinal metacentre above keel line
KM_T	Height of transverse metacentre above keel line
Length between perpendiculars (L_{PP})	This is the horizontal distance between the forward and after perpendiculars.
Length on the designed load water-line (L_{WL})	This is the length, as measured on the water-line of the ship when floating in still water in the loaded, or designed, condition.
Length overall (L_{OA})	This is the length measured from the extreme point forward to the

	extreme point aft.
Lightship weight	This equals the weight of an empty ship i.e. a ship without load. It is fully equipped and ready to proceed to sea, but with no crew, passengers, stores, fuel, water, or cargo on board. The boiler or boilers, however, are filled with water to their working level.
MCT_{1CM}	Moment to change trim 1 cm, $MCT_{1CM} = \frac{\Delta GM_L}{100\ L} \approx \frac{\Delta BM_L}{100L} = \frac{pI_L}{100L}$
Midship section	This is the transverse section of the ship amidships.
Base line	This represents the lowest extremity of the ship. At the point where this line cuts the midship section a horizontal line is drawn, and it is this line which acts as the datum, or base line, for all hydrostatic calculations. Normally, this is the underside of keel.
Midship section area coefficient(C_M)	This is the ratio of the immersed area of the midship section to the area of the circumscribing rectangle having a breadth equal to the breadth of the ship and a depth equal to the draught. ie: $C_M = A_M \div (B \times T)$ C_M values range from about 0.85 for fast ships to 0.99 for slow ships.
Net or register tonnage	This represents the tonnage of a ship after certain approved deductions, ie nonfreight earning spaces, have been made from gross tonnage. A *register ton* represents 100 cubic feet of volume.
Parallel middle body (L_P)	This is the length over which the midship section remains unchanged.
Prismatic coefficient (C_P)	This is the ratio of the volume of displacement of the ship to the volume of the circumscribing solid having a constant section equal to the immersed midship section area A_M, and a length equal to the L_{PP} i.e. $C_P = \nabla \div (A_M \times L)$ The Cp is a measure of the longitudinal distribution of displacement of the ship, and its value ranges from about 0.55 for fine ships to 0.85 for full ships.
Rise of floor	This is the amount by which the line of the outer bottom plating amidships rises above the base line, when continued to the moulded breadth lines at each side.
Sheer	This is the curvature given to the decks in the longitudinal direction, and is measured at any point by the difference between the height at side at that point and the height at side amidships.
Trim	This is the difference between the draughts forward and aft. If the draught forward is greater than the draught aft it is called trim by the head, or bow. If the draught aft is greater, it is called trim by the stern.
Tumble-home	This is the amount by which the midship section falls in from the half-breadth line at any particular depth.
Tonnes per centimetre (TPC)	This is the mass which must be added to, or deducted from, a ship in order to change its mean draught by 1 cm.
Water-plane area coefficient(C_{WP})	This is the ratio of the area of the water-plane to the area of the circumscribing rectangle having a length equal to the L_{PP} and a breadth equal to B. ie: $C_{WP} = A_W \div (L \times B)$ The range of values is from about 0.70 for a fine ship to 0.90 for a full ship.



Courtesy http://www.dynagen.co.za/eugene/hulls/terms.html

Exercise:

Visit these websites and get acquainted with more ship terms:

1. http://www.midwestconnection.com/glshpng/glossary.htm
2. http://www.scribd.com/doc/18008262/Ship-Terms-Glossary
3. http://cruises.about.com/od/cruiseglossary/Cruise_Ship_and_Nautical_Term_Glossary.htm
4. http://www.islandregister.com/terms.html
5. http://phrontistery.info/nautical.html